

# When Cross-View Helps Most: Conflict-Aware Disaster Triage Across Spatial Observation Regimes

Draft updated: 2026-04-30 16:15 CDT

Yifan Yang

Department of Geography  
Texas A&M University  
College Station, Texas, USA

Lei Zou

Department of Geography  
Texas A&M University  
College Station, Texas, USA

## Abstract

Cross-view disaster assessment combines ground-level and overhead imagery to infer building damage, yet most prior work is still evaluated primarily through average test performance. That perspective obscures the cases in which fusion is most valuable: samples for which the single-view predictors disagree. This study examines cross-view disaster triage through that conflict-centered lens on two paired datasets with distinct spatial observation regimes: an Eaton/Altadena wildfire dataset introduced in this study from property-centric inspection imagery and overhead patches, and the Hurricane Ian CVDIsaster benchmark introduced by Li et al. [4], which pairs panoramic ground imagery with overhead patches. We introduce a Conflict-Aware Evaluation protocol (CAE) that reports overall F1, conflict-subset F1 under a soft disagreement threshold, and the conflict gain of cross-view fusion over the best single-view model. CAE shows that fusion is most useful on evidence-conflict cases, but that both the source and the magnitude of the gain depend on the ground-view regime. In wildfire imagery, the ground view is the clearer single-view arbitrator. In hurricane settings, the stronger single view becomes split-sensitive rather than universally fixed, arguing for a more careful regime-aware interpretation than a simple disaster-name comparison. We further show that tile-level cross-view conflict density forms an unsupervised spatial damage signal in the wildfire dataset, and that a conflict-aware training objective improves wildfire overall F1 from 0.9473 to 0.9520 and Conflict-F1 from 0.6618 to 0.7208. Taken together, these results reframe cross-view disaster assessment as conflict-aware, regime-aware spatial reasoning rather than average-case multimodal classification.

## Keywords

disaster assessment, cross-view learning, multimodal fusion, GeoAI, conflict-aware evaluation

## 1 Introduction

Post-disaster building assessment increasingly relies on multiple visual modalities. Overhead imagery provides broad geographic coverage and stable spatial context, while ground-level imagery exposes facade condition, street-facing damage, and local structural cues. Cross-view models are intended to combine these complementary signals. In practice, however, most cross-view studies still ask only whether multimodal fusion improves average performance. That question is incomplete.

The more consequential question is when cross-view fusion actually matters. In real deployments, many samples are easy. A heavily damaged building may be obvious in both views, while

an intact building may be equally clear from both street and overhead imagery. Cross-view fusion becomes most valuable when the two modalities provide competing evidence. Those disagreement cases are exactly where a deployment team would most want an additional source of information or a more reliable fusion policy.

This work studies cross-view disaster triage from that perspective. We argue that cross-view value is fundamentally conflict-dependent and regime-dependent. It is conflict-dependent because the practical value of fusion is concentrated in cases where single-view models disagree. It is regime-dependent because the relative usefulness of ground and overhead evidence changes with the spatial observation regime of the ground-view imagery. A property-centric inspection photo and a panoramic environmental photo are both ground-view images, but they encode very different relationships to the target building.

We investigate this hypothesis on two paired disaster datasets with different provenance. The first is our contributed Eaton/Altadena wildfire dataset, built from property-centric inspection imagery paired with overhead patches. The second is the Hurricane Ian CVDIsaster benchmark introduced by Li et al. [4], built from panoramic ground imagery paired with overhead patches. These datasets do not merely differ by disaster type. More importantly, they differ by spatial observation regime: the wildfire ground view is usually tightly aligned with the target property, whereas the hurricane ground view often contains broader environmental context and weaker target alignment.

To study these settings, we introduce a Conflict-Aware Evaluation protocol (CAE). CAE reports: (1) standard overall F1 on the full test set, (2) Conflict-F1 on the subset of samples where the single-view models disagree under a soft probability-gap threshold, and (3) the conflict gain of cross-view fusion over the best single-view baseline. This protocol shifts the paper’s evaluation axis from average-case performance to conflict resolution.

Using CAE, we find that cross-view fusion is consistently strongest on disagreement cases, but the dominant single view changes with the evaluation setting. Wildfire is clearly street-dominant on conflict cases. Hurricane is more nuanced: the endpoint split is close to balanced, while the broader label sensitivity split becomes overhead-favoring. That difference matters because it prevents overclaiming a simplistic “wildfire versus hurricane” story and instead supports a more careful observation-regime interpretation.

We then show that the conflict-centered framing leads to two useful method extensions. First, a tile-level conflict density map, computed from the disagreement magnitude between street-only and remote-only predictors, becomes a significant unsupervised

spatial damage indicator in the wildfire dataset. Second, a conflict-aware training objective, ConflictFocalLoss, improves wildfire cross-view performance across multiple weighting settings, with the best run increasing test F1 from 0.9473 to 0.9520.

This study makes four contributions:

- (1) We introduce a Conflict-Aware Evaluation protocol for paired-view disaster triage, shifting evaluation from average-case F1 to conflict-centered reasoning.
- (2) We show that cross-view gain is regime-dependent: wildfire is clearly street-dominant, while hurricane is more split-sensitive and therefore requires more careful interpretation than a single aggregate claim.
- (3) We demonstrate that tile-level cross-view conflict density forms an unsupervised spatial damage map in the wildfire setting.
- (4) We show that conflict can be operationalized beyond evaluation: ConflictFocalLoss improves the hardest wildfire conflict cases, and an adaptive inference cascade preserves or improves full cross-view performance while routing only a minority of samples to paired-view inference.

The broader implication is that cross-view disaster intelligence should not be modeled as generic multimodal fusion. It should be understood as conflict-aware, regime-aware spatial reasoning.

## 2 Related Work

Research on post-disaster visual assessment has largely focused on single-view remote sensing benchmarks, especially satellite-only settings such as xBD [1]. Broader GeoAI and disaster-response syntheses also show how central Earth observation, multimodal data integration, and operational decision support have become in this area [5, 10, 22]. Recent wildfire-oriented remote-sensing studies further show how high-resolution imagery, vegetation-risk modeling, and AI-based burned-area prediction are being used to characterize wildfire exposure and landscape change [7, 8, 11]. Satellite-only disaster assessment work has also explored multi-temporal fusion and transformer-based fusion architectures for building-level damage inference [12, 13]. This literature has established strong baselines for damage classification and mapping, but it does not address how different visual viewpoints interact when they disagree.

A smaller body of work studies ground-view imagery for disaster assessment. These approaches recover local structural evidence that overhead imagery cannot directly observe, but ground-only models face their own limitations: narrow viewpoint, occlusion, non-uniform coverage, and dependence on how well the captured scene aligns with the target structure. Recent street-view damage assessment work has shown that strong pre-trained vision backbones can extract useful hyperlocal cues from repeated ground observations [19]. Earlier Google Street View-based damage assessment work already demonstrated the value of street-level evidence for post-hurricane analysis and longer-horizon recovery interpretation [6, 21]. Related arbitration-oriented work has also shown that hurricane damage inference from street-view imagery benefits from models that explicitly resolve disagreement among visual cues rather than collapse all evidence into a single undifferentiated score [17]. Other recent work has combined street-view imagery

with structured building and hazard attributes, showing that multimodal fusion can improve post-hurricane damage classification even before overhead cross-view reasoning is introduced [16]. Related flood-vulnerability work has also shown that high-resolution remote sensing and street-view imagery can be integrated for urban building assessment under another hazard regime [15]. Taken together, these studies establish that street-level disaster perception is information-rich, but they do not address how street and overhead evidence should be reconciled when they conflict.

Recent cross-view disaster work begins to bridge these modalities. The closest prior direction is CVDDisaster, which studies cross-view geolocalization and damage estimation for Hurricane Ian using paired street-view and very high-resolution satellite imagery [4]. Follow-up work extends this line toward probabilistic generative location awareness and triple-objective optimization for disaster-related VGI geolocalization [3, 20]. That work demonstrates that paired-view learning is feasible and useful, but it still evaluates mainly by aggregate classification performance. It does not separate easy cases from disagreement cases, and it does not ask whether the dominant evidence source changes across spatial observation regimes.

Another nearby line of work studies generation rather than classification. Satellite-to-Street synthesizes post-disaster street-level views from overhead imagery using generative vision models [18]. That problem is complementary: it asks whether overhead observations can be translated into plausible ground views, whereas this study asks when real ground and overhead observations should be fused, trusted, or treated as conflicting evidence.

More broadly, multimodal fusion in computer vision and geospatial learning often assumes that modalities should be fused symmetrically. In disaster settings, that assumption is fragile. A property-centric inspection image and a panoramic environmental image carry different target alignment, different spatial context, and different failure modes. Our work is closer in spirit to GeoAI studies that ask how spatial observation processes shape model behavior, rather than merely which architecture gives the highest average score.

The study differs from prior cross-view disaster work in three ways. First, it defines the conflict subset as the primary evaluation target. Second, it uses a cross-regime design to compare two observation regimes rather than merely two datasets. Third, it translates disagreement analysis into spatial and algorithmic outputs: a conflict-density map, a conflict-aware training loss, and an adaptive conflict-resolver view of cross-view inference.

## 3 Problem Definition and Datasets

### 3.1 Datasets and Observation Regimes

*Task.* We study binary disaster triage from paired images. Each example contains a ground-view image and a paired overhead patch, with a binary damage label. The goal is to predict whether the target building belongs to the positive or negative damage class under a specified split definition.

Formally, the evaluation set is

$$\mathcal{D} = \{(x_i^{(s)}, x_i^{(r)}, y_i)\}_{i=1}^N, \quad (1)$$

where  $x_i^{(s)}$  is the street- or ground-view image,  $x_i^{(r)}$  is the remote-view overhead patch, and  $y_i \in \{0, 1\}$  is the binary damage label. For a model  $m$ , let  $p_i^{(m)} \in [0, 1]$  denote the predicted probability of the positive class and  $\hat{y}_i^{(m)} = \mathbf{1}[p_i^{(m)} \geq 0.5]$  its hard decision.

We consider three model settings under a shared training protocol:

- **street\_only**: ground image only
- **remote\_only**: overhead image only
- **crossview**: paired ground plus overhead fusion

**Eaton/Altadena Wildfire Dataset.** This paper contributes the Eaton/Altadena wildfire dataset used in our experiments. The dataset pairs property-centric inspection imagery with overhead patches. The ground images are typically taken from close range and are visually anchored to the target property. This gives the dataset strong target alignment and makes it a natural setting to study whether local facade evidence helps resolve conflicts in overhead interpretation.

For the primary wildfire binary triage task, we use:

- $0$  = No Damage + Affected
- $1$  = Minor + Major + Destroyed

We also evaluate a stricter sensitivity split:

- $0$  = No Damage
- $1$  = Affected + Minor + Major + Destroyed

The wildfire dataset additionally contains spatial grouping metadata, which allows tile-level geographic analysis and map-based outputs. Here a *tile* denotes one fixed overhead-image patch, identified in the metadata by its remote-tile filename. Multiple property-level samples can share the same tile when they are paired with the same overhead patch. Tile-level analysis therefore means aggregating labels or predictions across all samples assigned to that overhead patch. In Figure 1, each tile is plotted at a representative geographic location derived from its paired sample coordinates; the square color shows the tile’s observed positive-label damage rate, and the center dot indicates the tile sample count.

Figure 1 places the wildfire benchmark in geographic context and shows the local tile footprint in the Altadena / Eaton Fire area. This context matters because the wildfire benchmark is neighborhood-scale and spatially compact, which is precisely what enables the tile-level spatial analysis later in the paper.

**Hurricane Ian Benchmark.** The hurricane benchmark is the Hurricane Ian CVDIsaster dataset introduced by Li et al. [4]. We use it as an external paired-view benchmark for cross-regime comparison. The benchmark pairs panoramic ground imagery with overhead patches. Unlike the wildfire dataset, the ground view here is not tightly centered on a single property. It often includes broader street context, neighboring structures, and environmental clutter. We therefore treat it as a panoramic or environment-centric observation regime. The benchmark does not provide the same tile-level geographic metadata needed for our local GIS footprint maps, so the tile-level spatial analysis in this paper is conducted on the contributed wildfire dataset.

For the endpoint binary setting, we use:

- $0$  = MinorDamage
- $1$  = SevereDamage

We also evaluate a broader sensitivity setting:

- $0$  = MinorDamage
- $1$  = ModerateDamage + SevereDamage

The original hurricane endpoint split showed object-level leakage, meaning that imagery associated with the same damaged object could appear across train and test partitions. To check whether the endpoint result survives a stricter partition, we therefore rebuild the same endpoint label definition as a *grouped clean* split, where samples are grouped by object identifier before splitting. In the result tables, “Hurricane endpoint” refers to the original endpoint split, while “Hurricane endpoint (grouped clean)” refers to this leakage-controlled rerun. It is not a third label definition; it is a cleaner split of the same MinorDamage-versus-SevereDamage hurricane task.

**Why This Comparison Matters.** These datasets give us more than a two-disaster benchmark. They instantiate two different spatial observation regimes:

- **building-centric / property-centric**
- **panoramic / environment-centric**

This regime distinction is central to the study. We do not assume cross-view fusion is equally valuable in all paired-view settings. Instead, we test whether its gain depends on how directly the ground-view image captures the target structure.

### 3.2 Conflict-Aware Evaluation Protocol

Average test F1 is necessary but insufficient. To isolate the value of cross-view fusion in the cases that matter most, we define a Conflict-Aware Evaluation protocol (CAE).

For each completed setting, CAE reports:

- (1) **Overall-F1**: standard F1 over the full test set.
- (2) **Conflict-F1**: F1 over the disagreement subset defined by  $|p_{\text{street}} - p_{\text{remote}}| > \tau$ .
- (3) **Delta\_view**: the gain of cross-view fusion over the better single-view model on the conflict subset.

In the main experiments, we use  $\tau = 0.1$ . This soft conflict definition is more robust than a hard binary disagreement test and yields usable conflict-set sizes for both datasets. Table 2 reports the setting-specific conflict counts under this definition. In the main text, we especially emphasize the wildfire sensitivity split and the hurricane moderate-plus-severe split because they provide the clearest conflict-centered comparisons; both have label-independence permutation-test significance below  $1e-4$ . We report the wildfire sensitivity split rather than the broader primary wildfire collapse because the stricter split creates a more informative disagreement subset for the CAE analysis.

To keep the table labels compact, “Wildfire sensitive” denotes the stricter wildfire split defined above, “Hurricane moderate+severe” denotes the broader hurricane positive class, and “Hurricane endpoint (grouped clean)” denotes the leakage-controlled rerun of the original hurricane endpoint task.

More precisely, the conflict subset is

$$C_\tau = \left\{ i : \left| p_i^{(s)} - p_i^{(r)} \right| > \tau \right\}, \quad (2)$$

where  $p_i^{(s)}$  and  $p_i^{(r)}$  are the positive-class probabilities from the street-only and remote-only models, respectively. For any model  $m$

## (a) Spatial Observation Regimes and Geographic Context

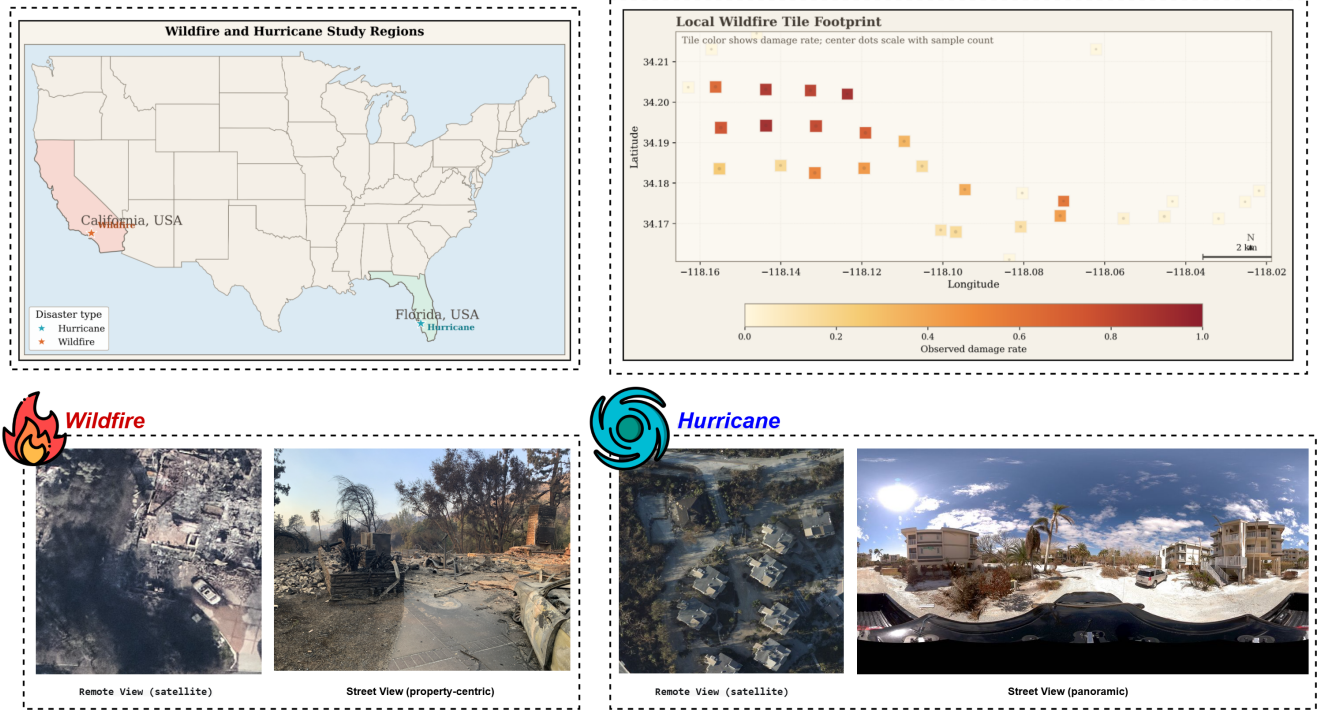


Figure 1: Overview of the study design and spatial observation regimes. The top row provides geographic context through the U.S. study-region locator map and the local wildfire tile footprint. In the footprint panel, each square represents one overhead-image tile used by one or more paired wildfire samples; square color encodes the observed positive-label damage rate within that tile, and the center dot scales with the number of samples assigned to the tile. The bottom row shows paired remote and ground views for the wildfire and hurricane settings, illustrating the property-centric wildfire regime and the panoramic hurricane regime considered throughout the paper.

and subset  $S \subseteq \{1, \dots, N\}$ , we compute

$$F1(m; S) = \frac{2 \text{Prec}(m; S) \text{Rec}(m; S)}{\text{Prec}(m; S) + \text{Rec}(m; S)}. \quad (3)$$

The conflict gain of cross-view fusion is then

$$\Delta_{\text{view}} = F1(\text{crossview}; C_r) - \max \{F1(\text{street}; C_r), F1(\text{remote}; C_r)\}. \quad (4)$$

Conceptually, CAE reframes the task. Instead of asking whether multimodal fusion improves average classification, CAE asks whether fusion helps when the single-view models disagree and which modality drives that improvement.

## 4 Methodology

### 4.1 Architecture, Evaluation Logic, and Conflict-Aware Extensions

**Base Triage Architecture.** We use a unified late-fusion architecture across all experiments. The `street_only` and `remote_only` models each use a single encoder followed by a binary classification head. The `crossview` model uses one encoder per view and builds a late-fusion representation from:

- the street embedding

- the overhead embedding
- their absolute difference
- their elementwise product

This architecture is intentionally simple. The goal is not to maximize raw architectural novelty, but to create a controlled setting in which we can study when and why paired-view fusion helps. The base implementation uses ResNet18 encoders [2], and we later compare against larger supervised and pretrained alternatives including CLIP [9].

Let  $f_s : \mathcal{X}^{(s)} \rightarrow \mathbb{R}^d$  and  $f_r : \mathcal{X}^{(r)} \rightarrow \mathbb{R}^d$  denote the street and remote encoders. The two view embeddings are

$$h_i^{(s)} = f_s(x_i^{(s)}), \quad h_i^{(r)} = f_r(x_i^{(r)}). \quad (5)$$

The late-fusion representation is

$$z_i = h_i^{(s)} \parallel h_i^{(r)} \parallel |h_i^{(s)} - h_i^{(r)}| \parallel (h_i^{(s)} \odot h_i^{(r)}), \quad (6)$$

where  $\odot$  denotes elementwise product. The cross-view classifier then predicts

$$p_i^{(\text{cv})} = \sigma(w^\top z_i + b), \quad (7)$$

with logistic sigmoid  $\sigma(\cdot)$ .

*View Dominance Switching.* Given CAE, we analyze conflict cases by asking which single-view model is more accurate within that subset. We also decompose conflicts into directional correction types:

- S→O: street is correct, overhead is wrong
- O→S: overhead is correct, street is wrong
- both wrong
- both correct under the soft disagreement threshold

This decomposition provides a mechanism-level explanation of cross-view gain. If one view more frequently corrects the other in a given regime, the fusion model should derive more of its conflict-resolution value from that modality.

We formalize the directional conflict subsets as

$$\mathcal{S}_{S \rightarrow O} = \left\{ i \in C_\tau : \hat{y}_i^{(s)} = y_i, \hat{y}_i^{(r)} \neq y_i \right\}, \quad (8)$$

$$\mathcal{S}_{O \rightarrow S} = \left\{ i \in C_\tau : \hat{y}_i^{(r)} = y_i, \hat{y}_i^{(s)} \neq y_i \right\}. \quad (9)$$

Their normalized frequencies,

$$\eta_{S \rightarrow O} = \frac{|\mathcal{S}_{S \rightarrow O}|}{|C_\tau|}, \quad \eta_{O \rightarrow S} = \frac{|\mathcal{S}_{O \rightarrow S}|}{|C_\tau|}, \quad (10)$$

act as a compact measure of which single view is the stronger conflict-set arbitrator.

*ConflictFocalLoss.* We introduce ConflictFocalLoss as a conflict-aware training objective for cross-view triage. The intuition is simple: disagreement-heavy samples are rarer than agreement cases, but they are also more important to the paper’s central problem. Standard binary loss treats every example equally. In contrast, ConflictFocalLoss increases the effective weight of samples whose street and overhead embeddings disagree more strongly.

We evaluate a gamma sweep over  $\{0.1, 0.25, 0.5, 1.0\}$ . The purpose of this sweep is not just to find a good point estimate, but to test whether conflict-aware weighting provides a stable positive signal across a range of strengths.

For sample  $i$ , we define a disagreement score

$$c_i = \left( \frac{1 - \cos(h_i^{(s)}, h_i^{(r)})}{2} \right)^\gamma, \quad (11)$$

where  $\gamma \geq 0$  controls how strongly high-disagreement examples are emphasized. The resulting training objective is

$$\mathcal{L}_{CF} = \frac{1}{B} \sum_{i=1}^B (1 + c_i) \text{BCE}(\ell_i, y_i), \quad (12)$$

where  $\ell_i$  is the cross-view logit and  $B$  is the minibatch size. When  $\gamma = 0$ , the weighting term collapses toward standard BCE; for moderate  $\gamma$ , the model puts more emphasis on samples whose two views disagree in feature space.

*Adaptive Inference Cascade.* We also study an adaptive two-stage cascade. Stage 1 uses the street-only and remote-only models. If their probabilities are close, we use their average as the prediction. If their disagreement exceeds a threshold, we route the sample to the full cross-view model as Stage 2. This design operationalizes the core claim: cross-view fusion is most useful as an on-demand conflict resolver, rather than an always-on inference path.

With routing threshold  $\delta$ , the cascade prediction is

$$p_i^{(\text{cascade})} = \begin{cases} \frac{1}{2} (p_i^{(s)} + p_i^{(r)}), & \text{if } |p_i^{(s)} - p_i^{(r)}| \leq \delta, \\ p_i^{(\text{cv})}, & \text{otherwise.} \end{cases} \quad (13)$$

This makes the conflict-routing policy explicit: stage two is invoked only on samples that exceed a disagreement budget.

*Conflict Density Spatial Map.* For the wildfire dataset, we convert paired-view disagreement into a tile-level spatial signal. For each geographic tile  $t$ , we compute:

$$\text{conflict\_density}(t) = \text{mean}_i |p_{\text{street},i} - p_{\text{remote},i}|$$

over all samples in the tile. This produces an unsupervised conflict-density map. If conflict density correlates with observed damage rate, then cross-view disagreement itself becomes a useful GIS-native spatial damage proxy.

Let  $\mathcal{I}(t)$  be the set of samples assigned to tile  $t$ . Then

$$\text{conflict\_density}(t) = \frac{1}{|\mathcal{I}(t)|} \sum_{i \in \mathcal{I}(t)} |p_i^{(s)} - p_i^{(r)}|, \quad (14)$$

and the corresponding empirical damage rate is

$$\text{damage\_rate}(t) = \frac{1}{|\mathcal{I}(t)|} \sum_{i \in \mathcal{I}(t)} y_i. \quad (15)$$

We then test the spatial-regime hypothesis using Spearman correlation between  $\text{conflict\_density}(t)$  and  $\text{damage\_rate}(t)$  over tiles.

## 4.2 Experimental Setup

We train all three base modes under a unified protocol and compare them across datasets, sensitivity settings, and backbone variants. The primary experiments use ResNet18 encoders. We also evaluate ResNet50, DINOv2, and CLIP in the cross-view setting to assess backbone dependence.

To evaluate robustness, we run:

- multi-seed experiments on wildfire and hurricane
- label-sensitivity experiments on both datasets
- a clean grouped rerun on hurricane
- threshold-sensitivity and permutation tests on the conflict subset

For mechanism analysis, we use a frozen semantic segmentation model, SegFormer-B5 [14], to estimate building occupancy and centrality in the ground-view imagery, which serves as a lightweight target-alignment proxy.

## 5 Results

### 5.1 Main Results

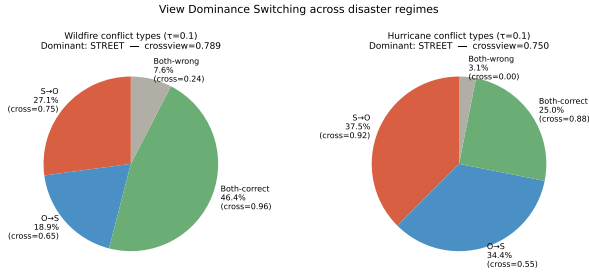
Across the original benchmark settings, cross-view fusion achieves the best overall test F1 in both disasters:

- wildfire: street\_only = 0.9604, remote\_only = 0.9653, crossview = 0.9713
- hurricane: street\_only = 0.8912, remote\_only = 0.9082, crossview = 0.9208

These numbers confirm that paired-view fusion is useful on average. The more informative result, however, lies in what happens on conflict cases, where CAE is more revealing than a single overall score.

**Table 1: View dominance switching analysis under the  $\tau = 0.1$  conflict definition.**  $n_c$  is the number of conflict samples.  $S \rightarrow O$  denotes street-correct / overhead-wrong conflicts, and  $O \rightarrow S$  denotes overhead-correct / street-wrong conflicts. The Dominant column names the stronger single-view model on the conflict subset, not necessarily the final best fused model.

| Split                     | $n_c$ | $S \rightarrow O$ | $O \rightarrow S$ | Dominant         |
|---------------------------|-------|-------------------|-------------------|------------------|
| Wildfire sensitive        | 435   | 0.2713            | 0.1885            | street           |
| Hurricane endpoint        | 32    | 0.3750            | 0.3438            | street, near-tie |
| Hurricane moderate+severe | 128   | 0.1094            | 0.1328            | overhead         |



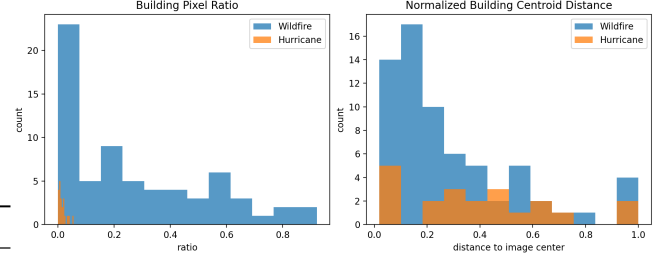
**Figure 2: Conflict-type decomposition under the completed  $\tau = 0.1$  view-dominance analysis.**

*CAE Makes the Hard Cases Visible.* Table 2 summarizes the full CAE view. Three patterns stand out. First, crossview remains strongest on the wildfire sensitive split and on the original hurricane endpoint setting under the CAE protocol. Second, the broader hurricane sensitivity setting still shows a positive but smaller  $\Delta \tau_{\text{view}}$ . Third, the grouped-clean hurricane endpoint split is more conservative: it remains a useful benchmark, but CAE shows that crossview is no longer dominant on the conflict subset there. This is exactly why CAE matters: it separates easy average performance from the harder disagreement cases that actually drive the mechanism story.

*View Dominance Switching Is Real but Split-Sensitive.* To move beyond scalar F1, we explicitly decompose conflict cases by directional correction type. Table 1 reports the  $\tau = 0.1$  breakdown across the wildfire sensitive split, the hurricane endpoint split, and the broader hurricane sensitivity split.

Wildfire still shows the clearest street-dominant conflict regime. On the hurricane endpoint split, however, the decomposition is much closer to balanced than the broader minor vs (moderate+severe) sensitivity split, where the overhead-correct / street-wrong share becomes larger and the rank order becomes crossview > remote > street. This means the study should avoid claiming a universal ‘hurricane equals overhead-dominant’ rule without naming the exact split being discussed. The stable statement is more precise: crossview stays strongest on the broader sensitivity split, while the identity of the stronger single view is split-dependent and therefore part of the phenomenon rather than noise.

*Alignment Proxy Supports the Regime Story.* The segmentation-based alignment proxy reinforces this argument. On the conflict



**Figure 3: A lightweight alignment proxy suggests much stronger target alignment in wildfire conflict images than in hurricane conflict images.**

subset, wildfire images have much higher building visibility and more central building evidence than hurricane images. Mean building ratio is 0.2684 for wildfire versus 0.0154 for hurricane, and mean center-building ratio is 0.4101 for wildfire versus 0.0271 for hurricane. This is exactly the regime distinction the study seeks to formalize: the property-centric wildfire ground view more often contains direct target evidence, whereas the hurricane ground view is much more environment-dominant.

## 5.2 Robustness and Regime Analysis

*The Conflict Story Persists Under Softer Thresholds.* The disagreement-centered story remains stable under the  $\tau = 0.1$  definition. On that conflict subset, wildfire accuracy is street = 0.7356, remote = 0.6529, and crossview = 0.7885. For the broader hurricane sensitivity setting, the corresponding numbers are street = 0.7031, remote = 0.7266, and crossview = 0.7500, with both permutation tests yielding  $p < 1e-4$ . This confirms that the disagreement-centered effect is not an artifact of a single hard-threshold definition.

*Backbone and Multi-Seed Robustness.* The backbone results show that larger or more generic pretraining is not automatically better for cross-view disaster triage. On wildfire crossview, best validation F1 is 0.9676 for ResNet18, 0.9693 for ResNet50, 0.9378 for DINOv2, and 0.7502 for CLIP. On hurricane crossview, the corresponding values are 0.9694, 0.9470, 0.7975, and 0.6437. These results strengthen the empirical case in two ways: the cross-view pattern is not a tiny-backbone artifact, and generic vision-language pretraining does not transfer cleanly to this task.

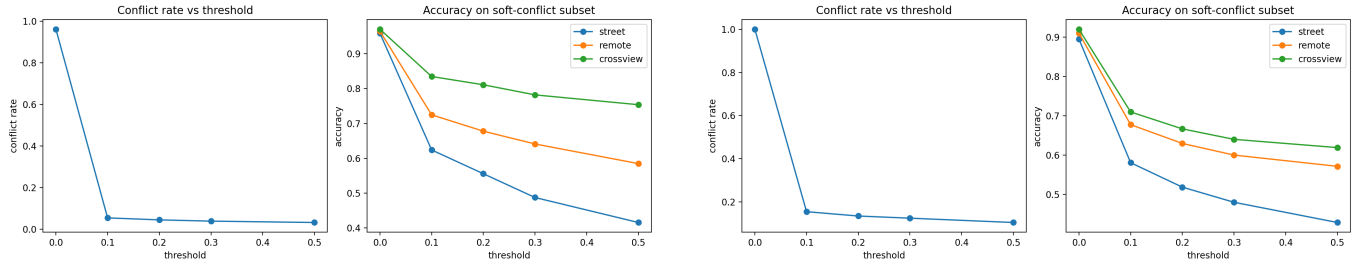
The main pattern is also stable across seeds. Wildfire multi-seed mean and standard deviation are  $0.9689 \pm 0.0011$  for crossview,  $0.9657 \pm 0.0001$  for street\_only, and  $0.9654 \pm 0.0015$  for remote\_only. Hurricane multi-seed values are  $0.9455 \pm 0.0063$ ,  $0.9433 \pm 0.0012$ , and  $0.8959 \pm 0.0065$ , respectively.

*Label Sensitivity and Clean Hurricane Control.* The broader hurricane binary mapping remains cross-view favorable: crossview = 0.8915, street\_only = 0.8738, and remote\_only = 0.8873. The broader wildfire binary mapping also preserves the ordering: crossview = 0.9473, street\_only = 0.9392, and remote\_only = 0.9238. These sensitivity checks matter because they show that the cross-view story is not limited to one exact label collapse.

After fixing object-level leakage in the hurricane endpoint split, we reran the core baselines on the same endpoint label definition

**Table 2: Conflict-Aware Evaluation (CAE) summary across completed settings. Wildfire rows report the stricter sensitivity split, which yields the most informative wildfire conflict subset for CAE. Hurricane endpoint (grouped clean) is the leakage-controlled grouped rerun of the same MinorDamage-versus-SevereDamage hurricane endpoint task. Conflict-F1 is computed only on samples where the street-only and remote-only predicted probabilities differ by more than  $\tau = 0.1$ . Crossview gain is reported on crossview rows as Conflict-F1(crossview) minus the larger Conflict-F1 of the two single-view baselines; dashes indicate quantities that are not applicable for that row.**

| Setting                            | Mode        | Overall-F1 | Conflict-F1 ( $\tau = 0.1$ ) | Crossview gain | $n_{\text{conflict}}$ |
|------------------------------------|-------------|------------|------------------------------|----------------|-----------------------|
| Wildfire sensitive                 | street_only | 0.9392     | 0.6230                       | –              | 435                   |
| Wildfire sensitive                 | remote_only | 0.9238     | 0.5237                       | –              | 435                   |
| Wildfire sensitive                 | crossview   | 0.9473     | 0.6618                       | +0.0388        | 435                   |
| Hurricane endpoint                 | street_only | 0.9055     | 0.6842                       | –              | 32                    |
| Hurricane endpoint                 | remote_only | 0.8969     | 0.5806                       | –              | 32                    |
| Hurricane endpoint                 | crossview   | 0.9192     | 0.7647                       | +0.0805        | 32                    |
| Hurricane moderate+severe          | street_only | 0.8738     | 0.7324                       | –              | 128                   |
| Hurricane moderate+severe          | remote_only | 0.8873     | 0.7742                       | –              | 128                   |
| Hurricane moderate+severe          | crossview   | 0.8915     | 0.7867                       | +0.0125        | 128                   |
| Hurricane endpoint (grouped clean) | street_only | 0.9016     | 0.8101                       | –              | 80                    |
| Hurricane endpoint (grouped clean) | remote_only | 0.8385     | 0.6526                       | –              | 80                    |
| Hurricane endpoint (grouped clean) | crossview   | 0.9008     | 0.7949                       | -0.0153        | 80                    |



**Figure 4: Threshold sensitivity under softer and harder conflict definitions. The conflict-aware story is not tied to a single threshold choice.**

using the grouped-clean partition. The resulting test F1 values are crossview = 0.9008, street\_only = 0.9016, and remote\_only = 0.8385. This is one of the most important sanity checks in the study because it prevents overclaiming. The clean split shows that the defensible claim is not “crossview always dominates.” Rather, crossview remains useful as a conflict-aware comparator and clearly outperforms remote-only on the clean split, while its advantage over street-only can shrink depending on the task definition and split rigor.

### 5.3 Method Extensions and Spatial Analysis

*ConflictFocalLoss, Adaptive Cascade, and Spatial Maps.* The complete ConflictFocalLoss sweep on wildfire sensitivity yields: baseline crossview = 0.9473, gamma = 0.1 gives 0.9514, gamma = 0.25 gives 0.9507, gamma = 0.5 gives 0.9520, and gamma = 1.0 gives 0.9470. Table 3 reports the corresponding overall and conflict-subset scores. Moderate conflict-aware weighting is consistently beneficial, while overly strong weighting collapses back toward baseline-level performance. The best setting improves the wildfire sensitivity task by +0.0047 F1. More importantly, the improvement is concentrated on the conflict subset: Conflict-F1 increases from 0.6618 to 0.7208

**Table 3: ConflictFocalLoss ablation on the wildfire sensitivity split. Conflict-F1 is computed on the  $\tau = 0.1$  disagreement subset, and  $\Delta$  Conflict-F1 is measured relative to the standard crossview baseline.**

| Setting            | Overall-F1 | Conflict-F1 | $\Delta$ Conflict-F1 |
|--------------------|------------|-------------|----------------------|
| Crossview baseline | 0.9473     | 0.6618      | –                    |
| $\gamma = 0.1$     | 0.9514     | 0.6906      | +0.0289              |
| $\gamma = 0.25$    | 0.9507     | 0.7190      | +0.0572              |
| $\gamma = 0.5$     | 0.9520     | 0.7208      | +0.0590              |
| $\gamma = 1.0$     | 0.9470     | 0.6957      | +0.0339              |

at gamma = 0.5, a gain of +0.0590 on the cases that CAE identifies as most consequential.

The adaptive inference cascade provides a complementary operational use of the same conflict principle. Instead of running cross-view fusion on every sample, the cascade first compares the street-only and remote-only predictors and only invokes the paired-view model when their probabilities differ by more than a threshold. Table 4 shows that this selective policy can match or exceed full

**Table 4: Adaptive inference cascade results.**  $\delta$  is the routing threshold on the absolute probability gap between the street-only and remote-only models. Stage-2 rate is the percentage of samples routed to the cross-view model.

| Split               | $\delta$ | Stage-2 | CV F1  | Casc. F1 |
|---------------------|----------|---------|--------|----------|
| Wildfire sens.      | 0.50     | 9.5%    | 0.9473 | 0.9498   |
| Hurricane mod.+sev. | 0.20     | 22.3%   | 0.8915 | 0.8990   |

cross-view F1 while calling the paired-view model for only a minority of samples.

On the wildfire dataset, tile-level conflict density also forms a significant unsupervised damage proxy:  $n\_tiles = 27$ , Spearman  $r = 0.5123$ , and  $p = 0.0063$ . This is a strong spatial-analysis result because it converts disagreement between two discriminative models into a spatial damage map without using tile-level labels at inference time.

## 6 Discussion and Limitations

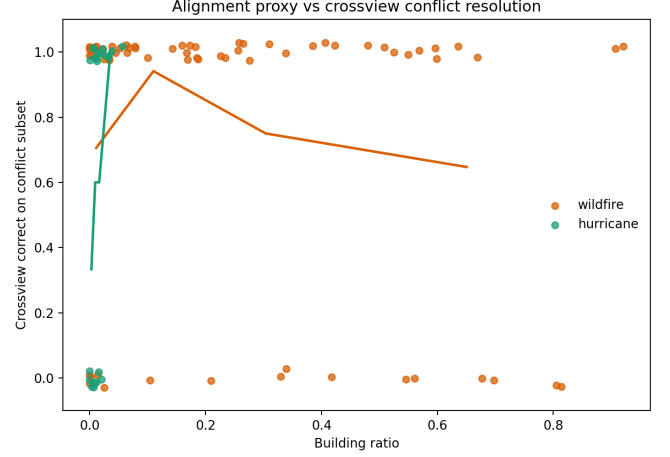
The results support a different way of thinking about cross-view fusion in disaster intelligence. The main value of fusion is not simply that it adds more information and nudges up average F1. Its more important function is to resolve evidence conflict. That is why conflict-centered evaluation is more revealing than average-case evaluation alone.

The results also suggest that “ground-view” is too coarse a category. A property-centric inspection image and a panoramic environmental view interact with overhead imagery in different ways. This is why the study uses the concept of spatial observation regime. In one regime, the ground view offers strong target-specific corrections. In another, the overhead image can remain the more trustworthy single-view source, or the balance can become split-sensitive depending on label construction and data leakage control.

This regime-centered explanation is especially relevant for geographic AI and remote sensing, but the paper’s claim is more specific than a generic GeoAI label. The central question is how the spatial alignment between the observer and the target changes the utility of each modality. That framing also creates room for adaptive inference policies, where expensive cross-view reasoning is reserved for the cases that actually need it.

The current study has several limitations. First, the spatial-map contribution is currently validated strongly for wildfire but not yet for hurricane, because the hurricane benchmark does not currently supply equally reliable tile-level geographic structure for the same analysis. We therefore treat wildfire as the primary GIS-native spatial analysis dataset and hurricane as a cross-regime comparison benchmark.

Second, the grouped-clean hurricane endpoint split leads to a more conservative result than the original endpoint split. This leakage-controlled split is a necessary boundary condition for the study: it shows that the argument should emphasize conflict-aware value and regime comparison rather than a blanket assertion that cross-view always dominates every baseline.



**Figure 5: Per-sample alignment correlation is weak, reinforcing that the alignment effect should be interpreted mainly at the regime level.**

Third, ConflictFocalLoss is currently validated only on the wildfire sensitive setting. The positive signal is encouraging, but broader validation remains future work.

## 7 Conclusion

The central claim is that cross-view disaster triage should be understood as conflict-aware, regime-aware spatial reasoning. Across wildfire and hurricane paired-view datasets, cross-view fusion is most useful on disagreement cases, not merely on average cases. The dominant single-view arbitrator changes across observation regimes and split constructions, which makes conflict-centered evaluation necessary for honest interpretation. We further show that conflict can be elevated from an analysis artifact to a method and spatial signal: it supports conflict-aware training, adaptive cross-view inference, and unsupervised tile-level damage mapping.

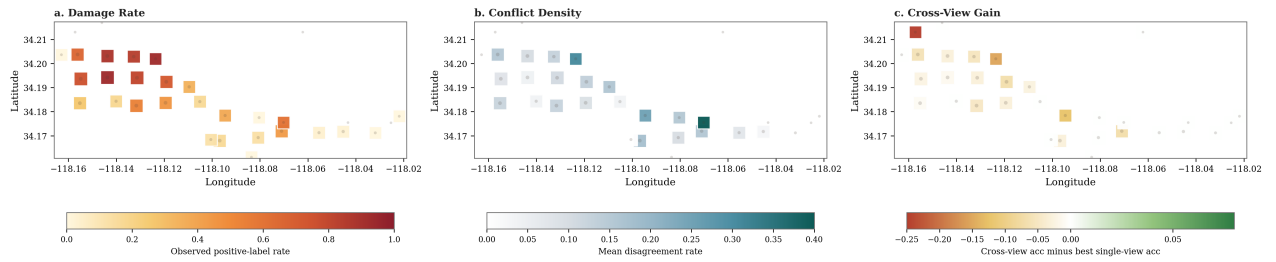
The broader takeaway is that multimodal disaster assessment should not ask only whether fusion helps. It should ask when fusion helps, which modality drives the gain, and what spatial regime makes that gain possible.

## Acknowledgments

The authors used GPT-based writing assistance only for grammar checking and language polishing of the author-written manuscript text. These tools were not used as sources of scientific evidence or to autonomously generate the study design, datasets, experimental results, figures, tables, interpretations, or conclusions. All analyses, visualizations, and manuscript content were reviewed, verified, and approved by the authors, who take full responsibility for the work.



**Figure 6: Qualitative conflict cases across wildfire and hurricane regimes. The panel illustrates how cross-view fusion resolves disagreement when the two single-view models provide competing evidence.**



**Figure 7: Large-format wildfire spatial maps reveal that high conflict density aligns with high realized damage and therefore supports a tile-level unsupervised spatial damage signal.**

## References

- [1] Ritwik Gupta, Bryce Goodman, Nirav Patel, Ricky Hosfelt, Sandra Sajeev, Eric Heim, Jigar Doshi, Keane Lucas, Howie Choset, and Matthew Gaston. 2019. Creating xBD: A Dataset for Assessing Building Damage from Satellite Imagery. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. 10–17.
- [2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. 770–778.
- [3] Hao Li, Fabian Deuser, Wenping Yin, Steffen Knoblauch, Wufan Zhao, Filip Biljecki, Yong Xue, and Wei Huang. 2026. Towards Generative Location Awareness for Disaster Response: A Probabilistic Cross-View Geolocalization Approach. *ISPRS Journal of Photogrammetry and Remote Sensing* 237 (2026), 130–145.
- [4] Hao Li, Fabian Deuser, Wenping Yin, Xuanshu Luo, Paul Walther, Gengchen Mai, Wei Huang, and Martin Werner. 2025. Cross-View Geolocalization and Disaster Mapping with Street-View and VHR Satellite Imagery: A Case Study of Hurricane IAN. *ISPRS Journal of Photogrammetry and Remote Sensing* 220 (2025), 841–854.

- [5] Z. Li, H. Li, Y. Yang, S. Wang, and Y. Zhu. 2025. Integrating Earth Observation Data into the Tri-Environmental Evaluation of the Economic Cost of Natural Disasters: A Case Study of 2025 LA Wildfire. *arXiv preprint arXiv:2505.01721* (2025). doi:10.48550/arXiv.2505.01721
- [6] Leslie Mabon. 2016. Charting Disaster Recovery via Google Street View: A Social Science Perspective on Challenges Raised by the Fukushima Nuclear Disaster. *International Journal of Disaster Risk Science* 7, 2 (2016), 175–185.
- [7] Srikanth A. Nagaraja, Istvan Kereszy, Chang Zhao, and Imre Bartos. 2025. From Vegetation to Vulnerability: Integrating Remote Sensing and AI to Combat Cheatgrass-Induced Wildfire Hazards in California. *Environmental and Sustainability Indicators* (2025), 100852.
- [8] Bo Pang, Sibo Cheng, Yuhang Huang, Yufang Jin, Yike Guo, I. Colin Prentice, Sandy P. Harrison, and Rossella Arcucci. 2025. Fire-Image-DenseNet (FIDN) for Predicting Wildfire Burnt Area Using Remote Sensing Data. *Computers & Geosciences* 195 (2025), 105783.
- [9] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning Transferable Visual Models from Natural Language Supervision. In *International Conference on Machine Learning*. PMLR, 8748–8763.
- [10] A. Raj, A. Shetgaonkar, L. Arora, D. Pradhan, S. S. Girija, and S. Kapoor. 2025. AI and Generative AI Transforming Disaster Management: A Survey of Damage Assessment and Response Techniques. In *2025 IEEE 49th Annual Computers, Software, and Applications Conference (COMPSAC)*. 1834–1840. doi:10.1109/COMPSAC65507.2025.00251
- [11] Yongli Tang and Chao Fan. 2025. Evaluating Building-Level Tree Cover Change in Southern California Wildland-Urban Interface Using High-Resolution Satellite Imagery. *Journal of Environmental Management* 380 (2025), 125160.
- [12] Ethan Weber and Hassan Kané. 2020. Building Disaster Damage Assessment in Satellite Imagery with Multi-Temporal Fusion. *arXiv preprint arXiv:2004.05525* (2020).
- [13] Weiwei Xiao, Jingyong Su, Yongyong Chen, and Guofeng Cao. 2023. Cross-Scale-Guided Fusion Transformer for Disaster Assessment Using Satellite Imagery. *IEEE Transactions on Geoscience and Remote Sensing* 61 (2023), 1–12.
- [14] Enze Xie, Wenhai Wang, Zhiding Yu, Anima Anandkumar, Jose M Alvarez, and Ping Luo. 2021. SegFormer: Simple and Efficient Design for Semantic Segmentation with Transformers. *Advances in Neural Information Processing Systems* 34, 12077–12090.
- [15] Ziyao Xing, Shuai Yang, Xuli Zan, Xinrui Dong, Yu Yao, Zhe Liu, and Xiaodong Zhang. 2023. Flood Vulnerability Assessment of Urban Buildings Based on Integrating High-Resolution Remote Sensing and Street View Images. *Sustainable Cities and Society* 92 (2023), 104467.
- [16] Zhuoqun Xue, Xiaojian Zhang, David O Prevatt, Jennifer Bridge, Susu Xu, and Xilei Zhao. 2024. Post-Hurricane Building Damage Assessment Using Street-View Imagery and Structured Data: A Multi-Modal Deep Learning Approach. *arXiv preprint arXiv:2404.07399* (2024).
- [17] Yifan Yang, Lei Zou, Wenjing Gong, Kani Fu, Zongrong Li, Siqin Wang, Bing Zhou, Heng Cai, and Hao Tian. 2026. DamageArbiter: A CLIP-Enhanced Multimodal Arbitration Framework for Hurricane Damage Assessment from Street-View Imagery. *arXiv preprint arXiv:2603.14837* (2026).
- [18] Yifan Yang, Lei Zou, and Wendy Jepson. 2026. Satellite-to-Street: Synthesizing Post-Disaster Views from Satellite Imagery via Generative Vision Models. *arXiv preprint arXiv:2603.20697* (2026).
- [19] Yifan Yang, Lei Zou, Bing Zhou, Daoyang Li, Binbin Lin, Joynal Abedin, and Mingzheng Yang. 2025. Hyperlocal Disaster Damage Assessment Using Bi-Temporal Street-View Imagery and Pre-Trained Vision Models. *Computers, Environment and Urban Systems* 121 (2025), 102335.
- [20] Wenping Yin, Fabian Deuser, Ziqi Liu, Jiaqi Wei, Xuanshu Luo, Martin Werner, Hao Li, and Yong Xue. 2026. Triple-Objective Cross-View Geolocalization of Disaster-Related VGI: The Case of Hurricane Ian. *International Journal of Geographical Information Science* 40, 1 (2026), 194–216.
- [21] Wei Zhai and Zhong-Ren Peng. 2020. Damage Assessment Using Google Street View: Evidence from Hurricane Michael in Mexico Beach, Florida. *Applied Geography* 123 (2020), 102252.
- [22] Lei Zou, Ali Mostafavi, Bing Zhou, Binbin Lin, Debjyoti Mandal, Mingzheng Yang, Joynal Abedin, and Heng Cai. 2023. GeoAI for Disaster Response. In *Handbook of Geospatial Artificial Intelligence*. CRC Press.